



AIM: Perform steep turns up to 60° AoB and manage turning upsets including spiral dive recovery.

P Departure/transit/return to aerodrome: Student as PIC; instructor may assist departure radio

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P HASSELL: student completes – **repeat every 15 minutes**

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DP 45° steep turn: Bank, Balance, Backpressure; overbanking tendency; alt ± 100 ft; rollout $\pm 10^\circ$; student practices L and R

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DP 60° steep turn: Same; increased backpressure vs 45°; student practices L and R

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DP Slipping turn: Patter: picture (turn rate apparent decrease), ball toward lower wing, aileron into turn; enter positively; above V_a ; student practices entry, hold, recovery

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D Skidding turn: Patter: picture (turn rate apparent increase), ball toward lower wing, aileron out of turn; temptation to pitch up; spin risk; conduct above V_a only; – **demo only; no student practice**

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DP Spiral dive: Bank increases, nose drops, airspeed builds; recovery – throttle, level wings, ease out; student recovers from instructor-initiated entry

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DP Dutch roll reversal: Coordinated rolls at 20° AoB alternating directions; student practices

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P Sustained 360° at 45°: Rollout on entry heading $\pm 10^\circ$

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P Sustained 360° at 60°: Rollout on entry heading $\pm 10^\circ$

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COMMON ERRORS

- **Altitude loss in steep turn** -> Bank correct first; then backpressure
- **Backpressure before wings level (spiral)** -> Level wings first – always
- **HASSELL** -> Before sequence; every 15 min – not required before each manoeuvre; maintain SA regarding traffic and airspace

STANDARDS FOR PROGRESSION

P S Progressing (P): 45° $\pm 10^\circ$, alt ± 150 ft; 60° attempted with guidance; spiral with minor prompting

S Solo standard (S): 60° turns $\pm 5^\circ$, alt ± 200 ft; spiral identified and recovered; HASSELL one prompt

C NC C standard: 60° turns $\pm 5^\circ$, alt ± 100 ft, ASI ± 10 kt, rollout $\pm 10^\circ$; spiral correct sequence; HASSELL every 15 min without prompting

⚠ SAFETY Above 2000 ft AGL: All recoveries above 1500 ft AGL · Spiral dive: confirm recovery sequence before entry · Ready to take control if bank $> 60^\circ$ or nose $> 30^\circ$ below horizon · Do not exceed V_A